

Can Frontier LLMs One-Shot Translate Lakota? (Mostly No.)

Evaluating Frontier LLM Translation Capability for Lakota

Lance Robertson · University of California, San Diego

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No frontier LLM reliably translates Lakota.

Even measuring how badly they fail is hard.

Lakḥótiyapi

the language under test

Why test this?

Lakḥótiyapi (Lakota) is a Siouan language UNESCO classifies as **critically endangered** — fewer than 2,000 first-language speakers, most over 65.

Today’s LLMs list translation support across dozens of languages, but providers don’t say what they can or can’t do per language. A reasonable user can’t tell Lakota apart from any of the others on the list — *if it does all these, why not this one?* Direct evaluation is the only way to find out.

Is it just memorized?

The evaluation set comes from a published dictionary, so contamination is the obvious worry. The Lakota source appears verbatim online for ~10% of pairs, but the Lakota and its English reference together for only 0.5%. Scores spread widely (6–61% equivalence), not clustered at ceiling — the signature of partial capability, not recall.

Setup

- **7 models.** Proprietary: Gemini 3.1 Pro, Claude Opus 4.6, Claude Sonnet 4.6, GPT-5.2. Open-weight: DeepSeek-V4-Pro, Qwen3.6-Plus, GLM-5.1.
- **200 sentence pairs** from the New Lakota Dictionary, conversational register, **both directions**.
- **Two conditions:** baseline vs. extended reasoning, where the API permits.
- **Metrics:** chrF + + and BLEU; diacritic-normalized chrF + +; an LLM *semantic* judge on the English side.

LLM-as-judge for L → E

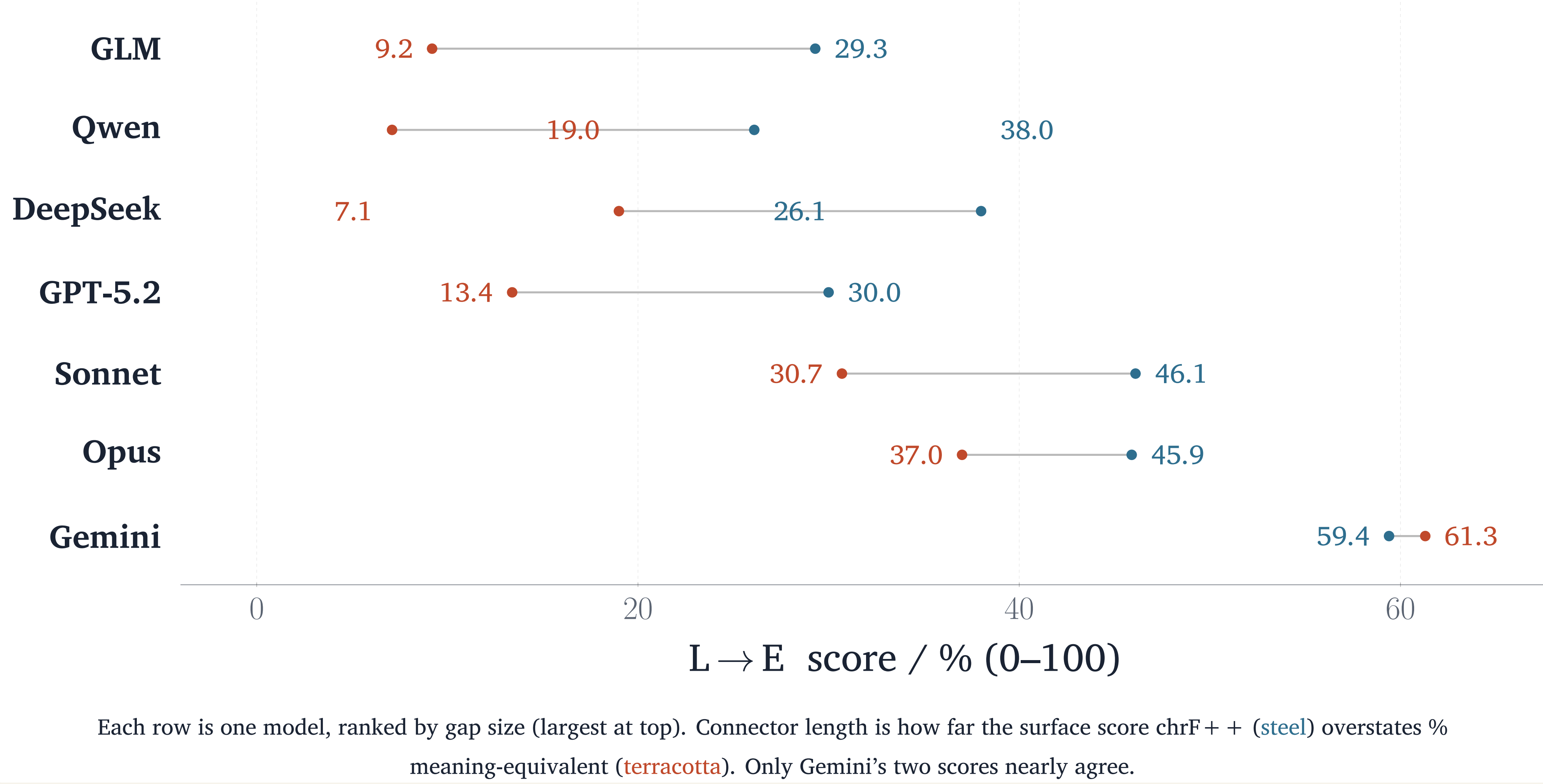
chrF + + measures character and word overlap, but it can reward plausible nonsense. We wanted an automated measure that could take meaning into account. We use an LLM-as-judge: for the L → E direction the output is English, so a model can assess whether each translation carries the reference’s meaning. To keep the judge from favoring its own family, every pair is scored by two LLMs from different families, Gemini 3 Flash and an independent Llama-3.3-70B. They agree substantially (Cohen’s $\kappa = 0.75, 0.89$ quadratic-weighted), so the judgments are cross-validated rather than one model’s unchecked read.

Standard Lakota Orthography

ḥ q ṭ u á ŋ ’

SLO encodes phonemic distinctions with diacritics; stripping them adds +3.3 to +6.5 chrF + + (E → L) — models get base characters right, marks wrong.

Surface metrics overstate capability



Translation quality (chrF++)

Model	L → E	E → L
<i>Proprietary</i>		
Gemini 3.1 Pro	59.4	42.6
Claude Opus 4.6	45.9	34.3
Claude Sonnet 4.6	46.1	27.0
GPT-5.2	30.0	23.6
<i>Open-weight</i>		
DeepSeek-V4-Pro	38.0	29.2
GLM-5.1	29.3	14.6
Qwen3.6-Plus	26.1	18.4

Best condition per model. Every model scores higher *out of* Lakota than *into* it.

How models fail

Reference	Model	Output	chrF++
We live in a small trailer house with no electricity.	DeepSeek	“small <i>sacred</i> tipi”	18.6
Make coffee (<i>Wakḥálya</i> yo).	DeepSeek	“Sanctify it!”	2.9
Did my daughter call?	GLM	“Did my younger brother come?”	27.6
Do you have a car?	GPT-5.2	“Do you have a horse?”	66.4

Idioms and coined terms are translated *element by element*: *Wakḥálya yo* (“make coffee,” literally “make it boil”) becomes “sanctify it.” A one-word near-miss — “car” rendered “horse” by a *frontier* model — still scores 66.4 chrF + +. Surface overlap and meaning preservation diverge.

Phrasebook concentration (L → E)

Model	Sc. 4	Sc. 5	Sc. 6	Δ
Gemini 3.1 Pro	52.7	58.2	81.9	+29.2
Claude Opus 4.6	41.6	41.0	65.6	+24.0
Claude Sonnet 4.6	40.3	45.1	64.1	+23.8
DeepSeek-V4-Pro	33.0	33.9	59.6	+26.6
GPT-5.2	27.9	23.7	45.7	+17.8
Qwen3.6-Plus	23.3	22.1	41.7	+18.4
GLM-5.1	27.3	26.0	41.0	+13.7

chrF + + stratified by NLD conversational score: 4 = more formal/complex, 6 = most conversational. Every model lifts sharply at 6 — performance concentrates on high-frequency phrasebook-like items rather than reflecting general capability.

Semantic equivalence (L → E)

Model	Equiv.	Partial	Not
Gemini 3.1 Pro	61.3	23.1	15.6
Claude Opus 4.6	37.0	21.0	42.0
Claude Sonnet 4.6	30.7	23.8	45.5
DeepSeek-V4-Pro	19.0	14.0	67.0
GPT-5.2	13.4	9.8	76.8
GLM-5.1	9.2	6.5	84.2
Qwen3.6-Plus	7.1	7.6	85.4

% of judged pairs, reasoning condition.

Takeaways

- No frontier LLM reliably translates Lakota, either direction (2026).
- chrF + + overstates low-resource capability — pair it with a meaning check.
- Open-weight doesn’t close the gap.

Extended reasoning changes refusal behavior more than quality (open-weight E → L: 0 → 36 refusals when enabled).

Origin

This evaluation grew out of a long-running interest in the possibility of crowdsourcing a Lakota training corpus. It started as a User Interface design problem: what could community-driven tools look like for gathering training data for indigenous-led AI? In a seminar, faculty gave us the provocation: that today’s LLMs are good enough you could just “*vibe-code*” a Lakota translator over a weekend. I wanted to investigate if this was feasible. My background in software development and LLM evaluations drove me to create this Lakota benchmark.



Data & code

github.com/robotson/
lakota-translation-benchmark